FOSS license scanning: The why, the how and the community approach

Caren Kresse
Open Source Automation Development Lab (OSADL) eG

https://www.osselot.org

- The **OSSelot** project provides a publicly available **database** with **curated compliance information** for frequently used FOSS components and related tooling.
- The database contains license and copyright analysis results:
  - README (and *info.json*) with metadata of the package, *e.g.* download location, purl, creator, reviews, comments
  - SPDX Tag:Value, JSON, YAML, RDF reports with concluded licenses, copyright notices, and comments on decisions per file
  - Disclosure document with aggregated license texts, copyright notices and acknowledgments per package

https://www.osselot.org

- Some manual input and review of the data is still indispensable, but:

The reuse of curated licensing and copyright information can tremendously reduce the time required to clear a software package!

Some dates and numbers

- The **OSSelot** project was publicly launched in the COOL lecture in December 2022.
- By now, there is a total of **90 packages** (541 unique) and 4,121,533 files analyzed.
- These packages are **over 90 % consistent** (→ all dependencies are also curated).
- **Some of the big ones**: gcc, llvm, e2fsprogs, 5 versions of the Linux kernel and 10 versions of zephyr are already curated.
- There are **19 contributors**.
- Several **external tools** (e.g. FOSSology, SCANOSS, Yocto) already integrate OSSelot material.

How to use OSSelot data

to clear a software package with a different version using FOSSology

General process

- Upload the OSSelot version of the package into FOSSology and **import the OSSelot data**.
  → **This clears the entire package automatically!**
- Run a **license scan** on the required version of the package.
- **Reuse** the result from the cleared OSSelot package on the required version.
- Only **new or modified** files remain to be cleared manually.

How to use OSSelot data

to clear a software package with a different version using FOSSology

Manually via GUI

- Find the **closest available version** of the required package in OSSelot, and **upload the source code** into FOSSology **without scanning**, but with **Reuse from OSSelot** referencing the given version. **This clears the package automatically!**
- Upload the source code of the required version into FOSSology with scanner runs and **Reuse from local** referencing the cleared OSSelot package.
- **Clear** the remaining files.

Automatable via FOSSology REST API

- Find out for which package versions **OSSelot data are available**.
- **Upload the source code** of the available OSSelot package into **FOSSology** (without scanning).
- Trigger the **OSSelot import** for the respective FOSSology upload. **This clears the package automatically!**
- Upload the source package of the required version into FOSSology, run **scanners** and **reuse**, referencing the corresponding OSSelot upload.
- **Clear** the remaining files (via GUI).

Home Search Browse Upload Jobs Organize Admin Help

fossology

Upload a New File

Version: [4.6.0.0], Branch: [master], Commit: [#b159d6] 2025/10/29 10:52 +01:00 built @ 2025/11/03 16:41 +01:00

To manage your own group permissions go into **Admin > Groups > Manage Group Users**. To manage permissions for this one upload, go to **Admin > Upload Permissions**.

This option permits uploading a single file (which may be iso, tar, rpm, jar, zip, bz2, msi, cab, etc.) from your computer to FOSSology. Your FOSSology server has imposed a maximum upload file size of 700Mbytes.

1. Select the folder for storing the uploaded files:

   Software Repository

2. Select the file(s) to upload:

   Browse... bash-5.2.37.tar.gz

[snip]

8. Select optional analysis:
   - [ ] Bucket Analysis
   - [ ] Compatibility License Analysis, scanning for licenses compatibility
   - [ ] Copyright/Email/URL/Author Analysis
   - [ ] ECC Analysis
   - [ ] IPRA Analysis
   - [ ] Keyword Analysis
   - [ ] MIME-type Analysis
   - [ ] Monk License Analysis
   - [ ] Nomos License Analysis
   - [ ] Ojo License Analysis
   - [ ] Package Analysis
   - [ ] REUSE.Software Analysis
   - [ ] SCANOSS Toolkit
   - [ ] Software Heritage Analysis

[snip]

10. (Optional) Reuse

    Set the reuse information

11. ScanCode Toolkit, scan for
    - [ ] License
    - [ ] Copyright
    - [ ] Email
    - [ ] URL

After you press Upload, please be patient while your file is transferring.

Upload

Reuse configuration

bash-5.2.37.tar.gz

- (Optional) Reuse

Reuse source: OSSelot SPDX2

Package name: bash Fetch Versions

Versions:
- ○ 5.1.16
- ○ 5.2.15
- ○ 5.2.32
- ◉ 5.2.37

Import options:

Create new licenses as:
- ◉ License candidate
- ○ New license

Match license using:
- ○ Shortname
- ◉ SPDX ID

Add the License Info as findings from:
- [ ] SPDX tag of type licenseInfoInFile
- [x] SPDX tag of type licenseConcluded

- [x] Add concluded licenses as decisions
  - [x] Also overwrite existing decisions
  - [ ] Import as "to be discussed"
- [ ] Add the copyright information as textfindings

Close

Home Search Browse Upload Jobs Organize Admin Help

License Browser

Version: [4.6.0.0], Branch: [master], Commit: [#b159d6] 2025/10/29 10:52 +01:00 built @ 2025/11/03 16:41 +01:00

Folder: Software Repository/
bash-5.2.37.tar.gz

Software Heritage | License Browser | File Browser | Spasht | Copyright | ECC | Email/URL/Author | IPRA | Keyword |

Display 50 files (tree view or flat)

Clear

	-- filter for scan results --	-- filter for edited results --	Open		-select option-
Files	Scanner Results (N: nomos, M: monk, Nk: ninka, I: reportImport, O: ojo, Sc: scancode, Sp: spasht, Rs: reso, So: scanoss)	Edited Results	Clearing Status	Cleared / Open / Total	Actions
bash-5.2.37.tar/ bash-5.2.37	Artistic-1.0-Perl, Autoconf-exception, Bison-exception, Bison-exception-2.2, BSD-3-Clause, BSD-4-Clause-UC, BSD-style, CC-BY-SA-3.0, CC-BY-SA-4.0, Freeware, FSF, FSFAP, FSFUL, FSFULLR, Gettext-Permission-Notice, GFDL, GFDL-1.1, GFDL-1.1-or-later, GFDL-1.3, GFDL-1.3-invariants-only, GFDL-1.3-no-invariants-or-later, GFDL-1.3-or-later, GNU-Manpages, GPL, GPL-1.0-or-later, GPL-2.0-only, GPL-2.0-or-later, GPL-2.0-or-later-WITH-Autoconf-exception, GPL-2.0--with-bison-exception, GPL-3.0-only, GPL-3.0-or-later, GPL-3.0-or-later-WITH-Autoconf-exception, GPL-3.0-or-later-WITH-Bison-exception, GPL-3.0-or-later-WITH-Bison-exception-2.2, GPL-3.0-or-later-WITH-TeX-exception, HPND, ISC, Latex2e, LGPL-2.0-or-later, LGPL-2.1-or-later, License-of-GNU-Licenses, LicenseRef-scancode-autoconf-simple-exception, LicenseRef-scancode-autoconf-simple-exception-2.0, LicenseRef-scancode-bison-exception-2.0, LicenseRef-scancode-free-unknown, LicenseRef-scancode-fsf-notice, LicenseRef-scancode-fsf-unlimited-no-warranty, LicenseRef-scancode-other-permissive, LicenseRef-scancode-public-domain, LicenseRef-scancode-tex-exception, LicenseRef-scancode-unknown-license-reference, LicenseRef-scancode-warranty-disclaimer, MIT, Non-commercial, NOT-public-domain, NTP, OFL-1.1, Public-domain, Public-domain-ref, See-file, See-file.COPYING, TeX-exception, UnclassifiedLicense, X11, X11-possibility	GPL-2.0-or-later, GFDL-1.3-no-invariants-or-later, GPL-3.0-or-later-WITH-Bison-exception, X11, GNU-Manpages, FSFULLR, License-of-GNU-Licenses, Gettext-Permission-Notice, GPL-2.0-or-later-WITH-Autoconf-exception, Public-domain, NTP, FSFUL, GPL-3.0-or-later-WITH-Bison-exception-2.2, Latex2e, FSF, FSFAP, BSD-4-Clause-UC, GPL-3.0-or-later, GPL-3.0-or-later-WITH-Autoconf-exception, GPL-3.0-or-later-WITH-TeX-exception	●	1042 / 1042 / 3319	[Tag][Edit][Bulk]

100 % cleared

Showing 1 to 1 of 1 files

Page 1 of 1

Home Search Browse Upload Jobs Organize Admin Help

fossology

Upload a New File

Version: [4.6.0.0], Branch: [master], Commit: [#b159d6] 2025/10/29 10:52 +01:00 built @ 2025/11/03 16:41 +01:00

To manage your own group permissions go into **Admin > Groups > Manage Group Users**. To manage permissions for this one upload, go to **Admin > Upload Permissions**.

This option permits uploading a single file (which may be iso, tar, rpm, jar, zip, bz2, msi, cab, etc.) from your computer to FOSSology. Your FOSSology server has imposed a maximum upload file size of 700Mbytes.

1. Select the folder for storing the uploaded files:

   Software Repository

2. Select the file(s) to upload:

   Browse... bash-5.3.tar.gz

[snip]

8. Select optional analysis:
   - [ ] Bucket Analysis
   - [ ] Compatibility License Analysis, scanning for licenses compatibility
   - [x] Copyright/Email/URL/Author Analysis
   - [ ] ECC Analysis
   - [ ] IPRA Analysis
   - [ ] Keyword Analysis
   - [ ] MIME-type Analysis
   - [x] Monk License Analysis
   - [x] Nomos License Analysis
   - [x] Ojo License Analysis
   - [ ] Package Analysis
   - [ ] REUSE.Software Analysis
   - [ ] SCANOSS Toolkit
   - [ ] Software Heritage Analysis

[snip]

10. (Optional) Reuse

    Set the reuse information

11. ScanCode Toolkit, scan for
    - [ ] License
    - [ ] Copyright
    - [ ] Email
    - [ ] URL

After you press Upload, please be patient while your file is transferring.

Upload

Reuse configuration

bash-5.3.tar.gz

- (Optional) Reuse

Reuse source: Local

- [x] Select an already uploaded package for reuse in specific folder Software Repository (fossy:5)
- [x] Enhanced reuse (slower)
- [x] Reuse main license/s
- [x] Reuse report configuration settings
- [x] Reuse deactivated copyrights

Upload to reuse:

× bash-5.2.37.tar.gz from 2025-11-03 18:13:02 (open)

Close

Home Search Browse Upload Jobs Organize Admin Help

fossology

License Browser

Version: [4.6.0.0], Branch: [master], Commit: [#b159d6] 2025/10/29 10:52 +01:00 built @ 2025/11/03 16:41 +01:00

Folder: Software Repository/
bash-5.3.tar.gz

Software Heritage | License Browser | File Browser | Spasht | Copyright | ECC | Email/URL/Author | IPRA | Keyword | Browse

Display 50 files (tree view or flat) Clear

	-- filter for scan results --	-- filter for edited results --	Open		-select option-
Files	Scanner Results (N: nomos, M: monk, Nk: ninka, I: reportImport, O: ojo, Sc: scancode, Sp: spasht, Rs: reso, So: scanoss)	Edited Results	Clearing Status	Cleared / Open / Total	Actions
bash-5.3.tar/ bash-5.3	Autoconf-exception, Autoconf-exception-generic, Autoconf-exception-generic-3.0, Bison-exception-2.2, BSD-4-Clause-UC, BSD-style, CC-BY-SA-3.0, Freeware, FSF, FSFAP, FSFUL, FSFULLR, GFDL, GFDL-1.3, GFDL-1.3-only, GFDL-1.3-or-later, GNU-Manpages, GPL, GPL-2.0-only, GPL-2.0-or-later, GPL-2.0+-with-bison-exception, GPL-3.0-only, GPL-3.0-or-later, ISC, LGPL-2.1-or-later, Linux-man-pages-copyleft-2-para, MIT, No_license_found, Non-commercial, NOT-public-domain, Public-domain, Public-domain-ref, See-file, See-file.COPYING, TeX-exception, Texinfo-exception, UnclassifiedLicense, X11, X11-possibility	GPL-2.0-or-later, GFDL-1.3-no-invariants-or-later, X11, FSFULLR, License-of-GNU-Licenses, Gettext-Permission-Notice, GPL-2.0-or-later-WITH-Autoconf-exception, Public-domain, FSFUL, FSF, GPL-3.0-or-later, GPL-3.0-or-later-WITH-TeX-exception	●	479 / 1151 / 3444	[Tag][Edit][Bulk]

41 % cleared

Showing 1 to 1 of 1 files

Page 1 of 1

- 41 % is comparatively low and likely due to the major version upgrade.
- Reusing bash-5.2.32 for bash-5.2.37 clears 94 %.

fossology

Home Search Browse Upload Jobs Organize Admin Help

License Browser

Version: [4.6.0.0], Branch: [master], Commit: [#b159d6] 2025/10/29 10:52 +01:00 built @ 2025/11/03 16:41 +01:00

Folder: Software Repository/
bash-5.3.tar.gz

Software Heritage | License Browser | File Browser | Spasht | Copyright | ECC | Email/URL/Author | IPRA | Keyword | Browse

Display 50 files (tree view or flat) Clear

	-- filter for scan results --	-- filter for edited results --	Open		-select option-
Files	Scanner Results (N: nomos, M: monk, Nk: ninka, I: reportImport, O: ojo, Sc: scancode, Sp: spasht, Rs: reso, So: scanoss)	Edited Results	Clearing Status	Cleared / Open / Total	Actions
bash-5.3.tar/ bash-5.3	Autoconf-exception, Autoconf-exception-generic, Autoconf-exception-generic-3.0, Bison-exception-2.2, BSD-4-Clause-UC, BSD-style, CC-BY-SA-3.0, Freeware, FSF, FSFAP, FSFUL, FSFULLR, GFDL, GFDL-1.3, GFDL-1.3-only, GFDL-1.3-or-later, GNU-Manpages, GPL, GPL-2.0-only, GPL-2.0-or-later, GPL-2.0+-with-bison-exception, GPL-3.0-only, GPL-3.0-or-later, ISC, LGPL-2.1-or-later, Linux-man-pages-copyleft-2-para, MIT, No_license_found, Non-commercial, NOT-public-domain, Public-domain, Public-domain-ref, See-file, See-file.COPYING, TeX-exception, Texinfo-exception, UnclassifiedLicense, X11, X11-possibility	GPL-2.0-or-later, GFDL-1.3-no-invariants-or-later, X11, FSFULLR, License-of-GNU-Licenses, Gettext-Permission-Notice, GPL-2.0-or-later-WITH-Autoconf-exception, Public-domain, FSFUL, FSF, GPL-3.0-or-later, GPL-3.0-or-later-WITH-TeX-exception	●	924 / 1151 / 3444	[Tag][Edit][Bulk]

Showing 1 to 1 of 1 files

Page 1 of 1

41 % cleared (two bulk scans bring this up to 80 %)

- 41 % is comparatively low and likely due to the major version upgrade.
- Reusing bash-5.2.32 for bash-5.2.37 clears 94 % (without additional bulk scans).

How to use OSSelot data

to clear a software package with a different version using FOSSology

Manually via GUI

- Find the **closest available version** of the required package in OSSelot, and **upload the source code** into FOSSology **without scanning**, but with **Reuse from OSSelot** referencing the given version. **This clears the package automatically!**
- Upload the source code of the required version into FOSSology with scanner runs and **Reuse from local** referencing the cleared OSSelot package.
- **Clear** the remaining files.

Automatable via FOSSology REST API

- Find out for which package versions **OSSelot data are available**.
- **Upload the source code** of the available OSSelot package into **FOSSology** (without scanning).
- Trigger the **OSSelot import** for the respective FOSSology upload. **This clears the package automatically!**
- Upload the source package of the required version into FOSSology, run **scanners** and **reuse**, referencing the corresponding OSSelot upload.
- **Clear** the remaining files (via GUI).

List available OSSelot data

constants

parameters

```
$ curl -L -X GET \
  "https://fossology-server.com/repo/api/v1/osselot/packages/bash/versions" \
  -H 'Authorization: Bearer [token]' | json_pp

[
   "5.2.37",
   "5.2.32",
   "5.2.21",
   "5.2.15",
   "5.1.16"
]
```

Upload OSSelot source code into FOSSology

```
$ curl -L -X POST \
  "https://fossology-server.com/repo/api/v1/uploads" \
  -H 'folderId: [parent folder id]' \
  -H 'uploadDescription: OSSelot Baseline' \
  -H 'uploadType: url' \
  -H 'Authorization: Bearer [token]' \
  -H 'Content-Type: application/json' \
  --data-raw '{"location": { \
    "url": "https://ftp.gnu.org/gnu/bash/bash-5.2.37.tar.gz", \
    "name": "bash-5.2.37" \
}}'
{"code":201,"message":9,"type":"INFO"}
```

upload id

Trigger OSselot reuse

upload id

```
$ curl -L -X POST \
  "https://fossology-server.com/repo/api/v1/uploads/9/osselot/import" \
  -H 'Authorization: Bearer [token]' \
  -H 'Content-Type: application/json' \
 --data-raw '{ "package": "bash", "version": "5.2.37",
      "options": {
            "addLicenseInfoFromInfoInFile": false,
            "addLicenseInfoFromConcluded": true,
            "addConcludedAsDecisions": true,
            "addConcludedAsDecisionsTBD": false
}}'
{"code":202,"message":"Import job scheduled successfully","type":"INFO","jobId":138}
```

Get clearing status of OSSelot package

upload id

```
$ curl -L -X GET \
  "https://fossology-server.com/repo/api/v1/uploads/9/summary" \
  -H 'Authorization: Bearer [token]' | json_pp | grep filesToBeCleared

  "filesToBeCleared" : 0,
```

Upload required source package into FOSSology

```
$ curl -L -X POST \
  "https://fossology-server.com/repo/api/v1/uploads" \
  -H 'folderId: [partent folder id]' \
  -H 'uploadDescription: my required package' \
  -H 'uploadType: file' \
  -H 'Authorization: Bearer [token]' \
  -H 'Content-Type: multipart/form-data' \
  -F 'fileInput=@bash-5.3.tar.gz;type=application/octet-stream'

{"code":201,"message":10,"type":"INFO"}
```

upload id

Run scans and reuse on required source package

```
$ curl -L -X POST \
  "https://fossology-server.com/repo/api/v1/jobs" \
  -H 'Authorization: Bearer [token]' \
  -H 'Content-Type: application/json' \
  -H 'folderId: [folder id]' \
  -H 'uploadId: 10' \
  --data-raw '{ \
    "analysis": {"copyright_email_author": true, "monk": true, "nomos": true,
                                                                  "ojo": true}, \
    "reuse": {"reuse_upload": 9, \
      "reuse_group": "[reuse group name]", "reuse_enhanced": true, \
      "reuse_main": true, "reuse_report": true, "reuse_copyright": true} \
  }'
```

upload id

reuse upload id

Get clearing status of required source package

upload id

```
$ curl -L -X GET \
  "https://fossology-server.com/repo/api/v1/uploads/10/clearing-progress" \
  -H 'Authorization: Bearer [token]' | json_pp
```

```
{
   "totalFilesCleared" : 479,
   "totalFilesOfInterest" : 1151
}
```

An average embedded Linux system ...

- ... is covered by OSSelot to around 50 %:

acpid, adduser, apache2, apparmor, apt, audispd-plugins, base-files, base-passwd, bash, bsdutils, ca-certificates, conmon, containernetworking-plugins, coreutils, cpio, cron, crun, cryptsetup, dash, dbus, debconf, debian-archive-keyring, debianutils, diffutils, dirmngr, dmidecode, dmsetup, dpkg, e2fsprogs, findutils, firewalld, fuse-overlayfs, fuse3, gcc-12-base, gir1.2-glib-2.0, gir1.2-nm-1.0, golang-github-containers-common, golang-github-containers-image, grep, gzip, hostname, ifupdown, init, iproute2, iptables, iputils-ping, isc-dhcp-client, klibc-utils, kmod, less, libacl1, libapr1, libaprutil1, libarchive13, libargon2-1, libassuan0, libattr1, libb2-1, libbluetooth3, libbpf1, libbrotli1, libbsd0, libbz2-1.0, libc-bin, libcap-ng0, libcap2, libconfig9, libcrypt1, libcurl3-gnutls, libdb5.3, libdebconfclient0, libduktape207, libedit2, libelf1, libexpat1, libffi8, libgcrypt20, libgdbm-compat4, libglib2.0-0, libgmp10, libgnutls30, libgpg-error0, libgpgme11, libgpiod2, libgssapi-krb5-2, libhogweed6, libicu72, libidn2-0, libjansson4, libjson-c5, libkeyutils1, libldap-2.5-0, liblocale-gettext-perl, liblua5.3-0, liblz4-1, liblzma5, libmd0, libmm-glib0, libmnl0, libncursesw6, libndp0, libnetfilter-conntrack3, libnewt0.52, libnfnetlink0, libnftables1, libnftnl11, libnghttp2-14, libnpth0, libnsl2, libp11-kit0, libpam-modules, libpam-systemd, libpcre2-8-0, libperl5.36, libpolkit-agent-1-0, libpopt0, libproc2-0, libpsl5, libpython3-stdlib, libpython3.11-minimal, libreadline8, librsync2, librtmp1, libsasl2-2, libseccomp2, libselinux1, libsemanage2, libsepol2, libslang2, libslirp0, libsqlite3-0, libssh2-1, libssl3, libsubid4, libswupdate0.1, libtasn1-6, libteamdctl0, libtext-charwidth-perl, libtext-iconv-perl, libtext-wrapi18n-perl, libtirpc-common, libtss2-esys-3.0.2-0, libunistring2, liburiparser1, libwebsockets17, libwrap0, libxml2, libxxhash0, libyajl2, libyaml-0-2, libzck1, libzstd1, linux-image-6.12.11, logrotate, mawk, media-types, minicoredumper, nano, net-tools, ntpsec, pinentry-curses, podman, python3, sed, sensible-utils, skopeo, swupdate, sysuser-helper, sysvinit-utils, tar, tzdata, usr-is-merged, vim-common, watchdog, xml-core, zlib

Questions or feedback ...

... please get in touch:

info@osselot.org

C.Kresse@osadl.org

osadl.org/Contact